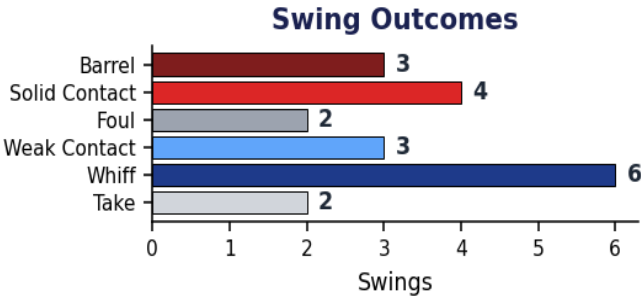
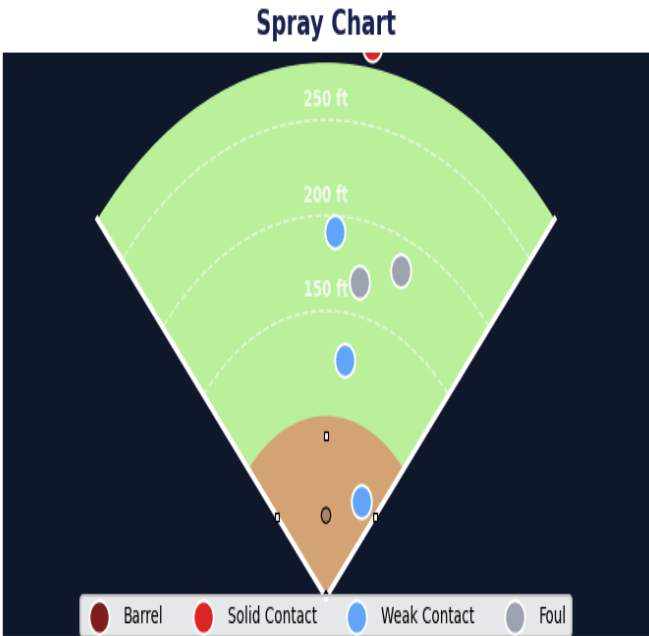


Lily Softball

■ Softball · Right-handed hitter · 2026 · Session: May 28, 2026

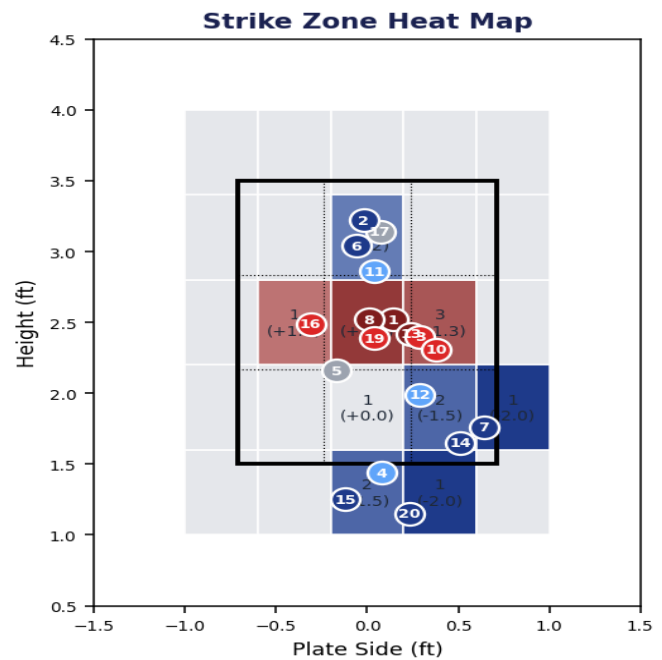
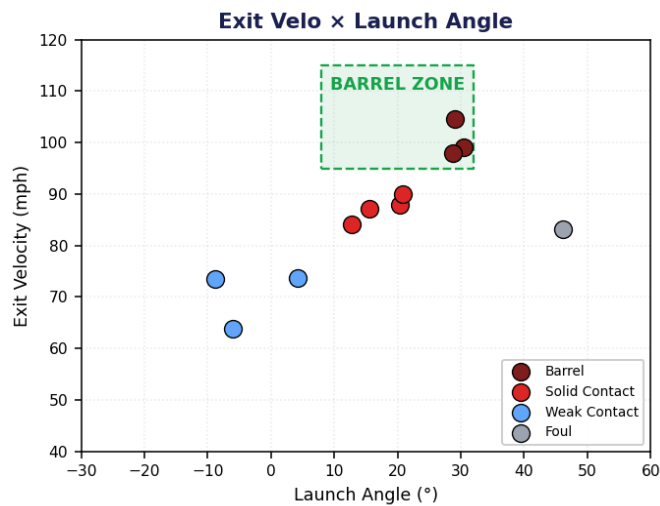
Swings	Avg Exit Velo	Peak Exit Velo	Avg Bat Speed	Barrel %	Whiff %
18	84.9 mph	104.6 mph	69.6 mph	16.7%	33.3%

Spray Chart & Swing Outcomes



Spray chart: every ball-in-play, colored by contact quality. Barrels (dark red) drive extra-base hits; weak contact (light blue) usually = routine outs.

Contact Quality & Zone Tendencies



Left: EV × Launch Angle. The shaded green box is the barrel zone (EV 95+ mph, LA 8°-32°) — where extra-base hits live. Right: the 5×5 strike-zone heat map colored by contact quality. Red zones = punishes pitches here; blue zones = struggles.

Performance by Pitch Type Faced

Pitch Type	Seen	Swings	Whiffs	Barrels	Avg EV	Avg LA
Change-Up	2	1	0	0	63.8 mph	-6.1°
Drop Ball	4	3	2	0	73.6 mph	4.2°
Rise Ball	4	4	2	0	78.3 mph	18.7°
Screwball	2	2	2	0	—	—
Softball Fastball	8	8	0	3	90.7 mph	28.0°

Swing Mechanics Critique

What's Working	Areas to Improve
✓ Strong bat speed. Avg bat speed was 69.6 mph — in the upper range for Softball (65-72 mph). <i>Drives exit velocity. Every +1 mph of bat speed ≈ +1.2 mph of exit velo.</i> ✓ Strong hip-shoulder separation. Peak hip-shoulder separation averaged 38.7° — in the upper range for Softball (38-48°). <i>Rubber-band torque drives bat speed and exit velocity.</i>	Clean swing — no corrections flagged.

Lily Softball — Action Plan

■ Today — Cooldown (within 30 min of finishing)

Standard Swing Session Cooldown [Recovery]

Drill: Wrist/forearm mobility + shoulder blade slides + light foam roll

Protocol: Forearm flexor stretches 3×30s each side. Scap slides 2×10. Foam roll T-spine 2 min.

Hitters accumulate forearm and lat fatigue from BP. Flushing the tissue speeds recovery for tomorrow.

■ **Forearm Stretch — Arm Stretch For Baseball** — YouTube

Triggered by: Standard cooldown after any swing session.

■ This Week — Before Next BP

Maintain and repeat what's working.

Questions or want to discuss next steps? This report was generated by Diamond Sports Lab — a coach-portable hitting + pitching analytics platform that fuses bat-flight, swing-mechanics, and contact-quality data into one workflow.

Full Hitting Drill Library — Reference

Every drill in the system, organized by the issue it targets. Pick whichever fits the hitter and your facility.

Bat Speed

When average bat speed is below the target for the hitter's level.

• Overload / Underload Bat Training

Drill: Alternate swings with a +10oz heavy bat and a -3oz light bat

Protocol: 3 sets × 5 heavy-bat swings, then 3 sets × 5 light-bat swings. 3 days this week.

Driveline-style protocol — reliably adds 2-4 mph of bat speed in 4-6 weeks by training fast-twitch fiber recruitment.

■ How to Train Bat Speed In The Cage

• Rotational Med-Ball Throws

Drill: Side-toss a 6-8 lb medicine ball into a wall, mimicking the swing turn

Protocol: 3 sets × 8 throws per side, 3 days this week. Drive from the back hip through the front side.

Builds raw rotational power that transfers directly to bat speed. Hitting starts in the legs and core.

■ Top 10 Med-Ball Drills to Increase Bat Speed & Power

• Resistance-Band Bat Speed Trainer

Drill: Dry swings against a hip-anchored resistance band

Protocol: 3 sets × 8 swings with band, then 3 sets × 5 free swings immediately after. 3 days this week.

Post-activation potentiation — heavy band loads the swing, then free swings feel explosive. Builds top-end speed.

■ Top 3 Resistance-Band Hitting Drills

• Top-Hand-Only Tee Drill

Drill: Choke up, top hand only, drive through the zone

Protocol: 2 sets × 10 reps, 3 days this week. Focus on snapping the wrist through contact.

Isolates the top hand — the dominant driver of bat speed late in the swing. Strengthens the snap.

■ Top-Hand Progression Drill

Hip-Shoulder Separation

When the hips and shoulders are firing together — no rubber-band torque.

• Coil-and-Hold Tee Drill

Drill: Load into back hip, pause 1 second, then unwind

Protocol: 3 sets × 5 reps, 3 days this week. The pause forces hip-shoulder dissociation.

Separation > 40° = rubber-band torque. Top HS hitters average 42-52°. Each 5° ≈ 1.5 mph bat speed.

■ Simple Hip Coil and Hold

- **Step-Behind Rotational Swing**

Drill: Stride foot crosses BEHIND the back foot, then unwinds into the swing

Protocol: 3 sets × 6 reps, 3 days this week. Off a tee — emphasizes hip lead.

Cross-step exaggerates the hip turn ahead of the shoulders, training proper sequencing.

- **Step Backs — Hitting Drill**

- **X-Drill (Hips vs. Shoulders)**

Drill: Hold a bat across shoulders, rotate hips one way, shoulders the other

Protocol: 3 sets × 8 controlled reps before BP, 3 days this week.

Trains the brain to feel hips and shoulders moving independently — the prerequisite for any rotational power.

- **Hip to Shoulder Separation — Baseball Hitting Mechanics**

Flat / Chopping-Down Swing Path

When the bat plane is below the productive 8°-16° attack-angle range.

- **High-Tee Launch Path**

Drill: Tee set at chest height — force the bat to launch upward

Protocol: 3 sets × 8 reps, 3 days this week. Aim for the upper third of the net.

Average MLB pitch enters the zone at -6°. Matching with +8-16° attack angle is the barrel-creation range.

- **High Tee — Hitting Drill**

- **PVC Stick Path Drill**

Drill: PVC pipe held at the slot — hitter must swing UNDER it on the way up

Protocol: 20 dry swings per set, 3 sets per session, 3 days this week.

Physical barrier teaches the body to drop the back shoulder slightly and work upward through contact.

- **PVC Hitting Drills To Help Improve Your Swing**

- **Pause-at-Launch Tee Drill**

Drill: Tee swings with a 1-second pause at the launch position

Protocol: 3 sets × 5 reps, 3 days this week. Hold the back-elbow slot, then explode upward.

Builds the feel of the back shoulder dropping into a positive attack angle before commitment.

- **Stride-Pause-Swing Tee Drill**

Steep Uppercut Swing Path

When attack angle is above 20° — whiffs and pop-ups.

- **Low/Middle Tee Mix**

Drill: 3 swings low tee, 3 swings middle tee, repeat

Protocol: 3 sets × 6 reps, 3 days this week. Forces the swing to flatten on low pitches.

Above 20° attack angle = whiffs and pop-ups. Mixing low and middle tees pulls the swing back into 8-16° range.

- **Low Tee — Hitting Drill**

• Top-Hand Path Correction

Drill: Top-hand-only swings on a mid-height tee, hands stay above the ball

Protocol: 2 sets × 10 reps, 3 days this week.

An uppercut usually comes from top-hand dropping early. This trains the hands to stay above the ball longer.

■ Drive the Ball More Consistently — Top Hand Drill

• Outside-Half Front Toss

Drill: Front toss on the outer third — drive everything oppo, gap to gap

Protocol: 2 sets × 15 swings, 2 days this week.

Hitting outside pitches oppo forces a flatter, longer bat path. Eliminates the steep pull-side cut.

■ Off-Set Front Toss Drill

Off-Plane Bat Path

When the bat is in the hitting zone for too small a window.

• Plyo Ball Into Uphill Net

Drill: Hitting plyo ball into a slight uphill angle, stay behind the ball

Protocol: 3 sets × 6 reps, 3 days this week.

On-plane time > 75% means the bat is in the hitting zone long enough to absorb timing mistakes.

■ Overloaded Plyo Ball Indoor Drill for Hitting

• Two-Tee Path Drill

Drill: Set TWO tees in a line — bat must clip both during the swing

Protocol: 3 sets × 5 reps, 3 days this week. Adjust the spacing as the path improves.

Physical waypoints force the bat to stay flat through the zone, not chop in and out.

■ Two Tee — Swing Path Drill

• Connection Band Drill

Drill: Resistance band looped around back elbow + lead hip — swing keeps connection

Protocol: 3 sets × 8 reps, 3 days this week.

Loss of connection = bat detaches from the body and the path goes off-plane. Band reinforces the link.

■ 3 Best Connection-Ball Drills

Slow Time-to-Contact

When the swing takes too long to get the barrel through the zone.

• Short-Bat Tee Work

Drill: Choke up on a bat, take 20 quick connected swings

Protocol: 3 sets per session, 3 days this week.

Faster TTC lets you wait longer on the pitch — fewer chase swings. Sub-0.16s separates HS varsity from elite.

■ Short Bat Drill (Window Drill)

- **Heavy-Bat Quick Swings**

Drill: Heavy bat (+10oz), short choked-up swings, max intent

Protocol: 2 sets × 8 reps, 2 days this week.

Heavy bat trains explosive power off the back hip, then transfers to a quicker free-swing.

- **Bat Speed and Quick Hands Drill**

- **Connection Ball Drill**

Drill: Soft ball pinned between back elbow and torso during swings

Protocol: 3 sets × 6 reps, 3 days this week. Ball must stay until contact.

Loss of connection adds time to contact. Ball forces the hands to stay tight, shortening the path.

- **Connection Ball Hitting Drills**

Elevated Whiff Rate

When the hitter is chasing offspeed or expanding the zone.

- **Sit-Fastball Drill**

Drill: 30 pitches — only swing at fastballs middle/outer-half. Take everything else.

Protocol: 2 days this week. Use a machine or live arm.

Whiffs > 35% almost always = chasing offspeed out of the zone. Trains the recognize-and-pass instinct.

- **Pitch Selection & Plate Discipline — Timing at Home Plate**

- **Bottle-Cap Vision Drill**

Drill: Front-toss with painted bottle caps — call color (red/blue) before swinging

Protocol: 2 sets × 15 reps, 2 days this week.

Forces tracking the ball into the zone before committing the swing. Builds late-recognition reflex.

- **Vision Training Baseball Batting Drill**

- **Multi-Color Ball Recognition**

Drill: Mix balls of different colors — coach calls the color, hitter only swings at that one

Protocol: 20 pitches × 2 sets, 2 days this week.

Trains pitch recognition under stress. Eliminates committing to a swing before the ball is identified.

- **Evan Longoria — Advanced Pitch Recognition Drill**

Weak Contact Pattern

When the hitter rolls over or pops up too many balls.

- **Front-Toss Away-Side BP**

Drill: Front toss on the outer third — drive everything oppo

Protocol: 2 sets × 20 reps, 2 days this week.

Weak contact = early commitment to pull. Going oppo trains keeping the bat behind the ball longer.

- **3 Tips To Hit For More Opposite Field Power**

- **Bottom-Hand-Only Drill**

Drill: Bottom hand only, choked up, drive line drives to centerfield

Protocol: 2 sets × 10 reps, 3 days this week.

Strengthens the lead hand's pull through contact. Weak bottom hand = bat slows at impact.

- **Bottom Hand Iso Drill — Knob Direction**

- **Inside-Outside Tee Series**

Drill: Alternate tee positions: inner third → middle → outer third

Protocol: 3 swings per location, 3 cycles per set, 2 sets per session, 3 days this week.

Builds adjustability — most weak contact comes from one path that only works on a middle pitch.

- **Inside/Outside Hitting Drill — Recognition & Adjustment**